Since 2009
AT KOLAR

KRISHNA CLASSES

WORKSHEET

SUBJECT- MATHS

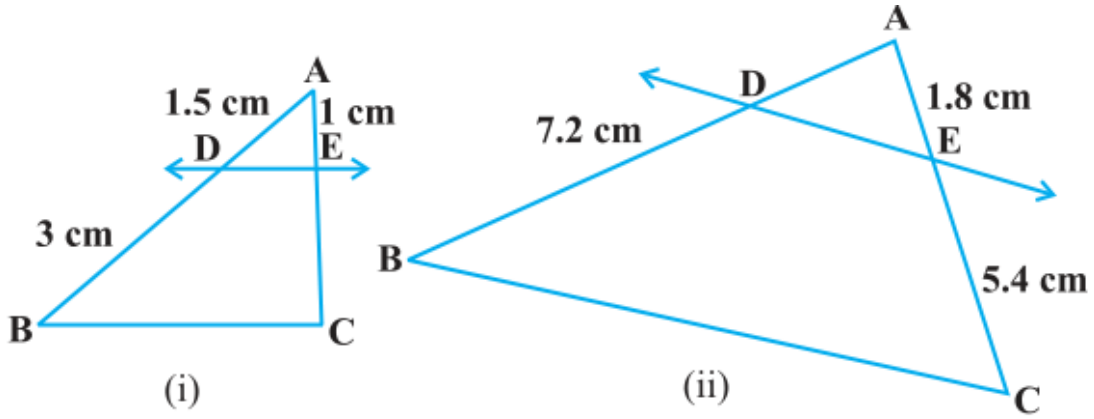
CHAPTER- Triangle

NAME OF STUDENT –

ISSUE DATE –

Level-1

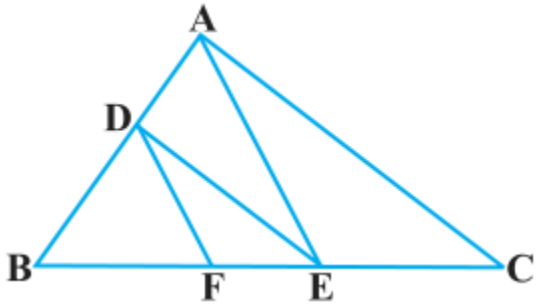
1. In figure. (i) and (ii), DE || BC. Find EC in (i) and AD in (ii).



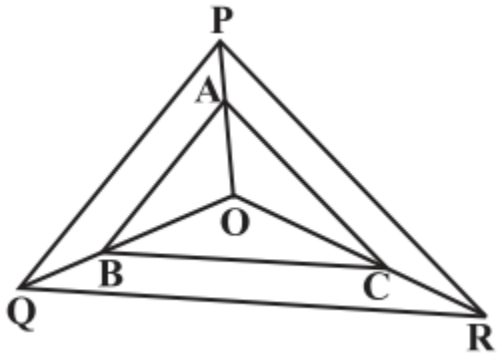
2. E and F are points on the sides PQ and PR respectively of a ΔPQR. For each of the following cases, state whether EF || QR.
   (i) PE = 3.9 cm, EQ = 3 cm, PF = 3.6 cm and FR = 2.4 cm

   (ii) PE = 4 cm, QE = 4.5 cm, PF = 8 cm and RF = 9 cm
   (iii) PQ = 1.28 cm, PR = 2.56 cm, PE = 0.18 cm and PF = 0.63 cm

3. In the figure, DE||AC and DF||AE. Prove that BF/FE = BE/EC



4. In the figure, A, B and C are points on OP, OQ and OR respectively such that AB || PQ and AC || PR. Show that BC || QR.



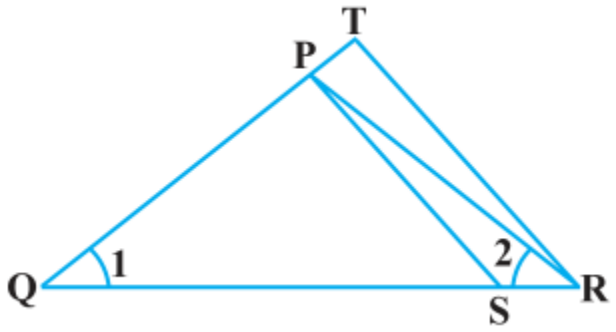
5. Using Basic proportionality theorem, prove that a line drawn through the mid-points of one side of a triangle parallel to another side bisects the third side. (Recall that you have proved it in Class IX).

6. ABCD is a trapezium in which AB || DC and its diagonals intersect each other at the point O. Show that AO/BO = CO/DO.

7. The diagonals of a quadrilateral ABCD intersect each other at the point O such that AO/BO = CO/DO. Show that ABCD is a trapezium.

8. Diagonals AC and BD of a trapezium ABCD with AB || DC intersect each other at the point O. Using a similarity criterion for two triangles, show that AO/OC = OB/OD

9. In the fig.6.36, QR/QS = QT/PR and ∠1 = ∠2. Show that ΔPQS ~ ΔTQR.



10. S and T are point on sides PR and QR of ΔPQR such that ∠P = ∠RTS. Show that ΔRPQ ~ ΔRTS.

11. E is a point on the side AD produced of a parallelogram ABCD and BE intersects CD at F. Show that ΔABE ~ ΔCFB.

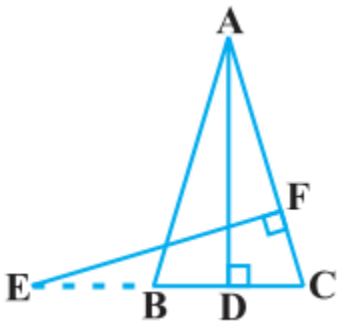
12. CD and GH are respectively the bisectors of ∠ACB and ∠EGF such that D and H lie on sides AB and FE of ΔABC and ΔEFG respectively. If ΔABC ~ ΔFEG, Show that:

    i. CD/GH = AC/FG

    ii. ΔDCB ~ ΔHGE

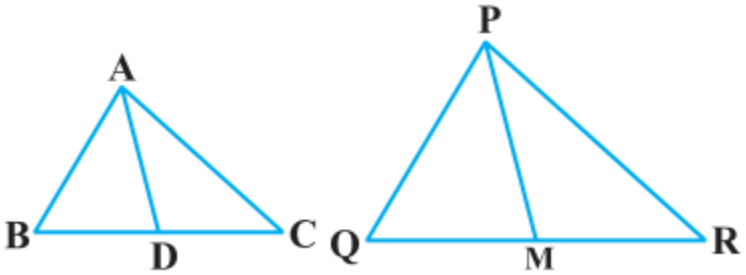
    iii. ΔDCA ~ ΔHGF

13. In the following figure, E is a point on side CB produced of an isosceles triangle ABC with AB = AC. If AD ⊥ BC and EF ⊥ AC, prove that ΔABD ~ ΔECF.



Level-2

14. Sides AB and BC and median AD of a triangle ABC are respectively proportional to sides PQ and QR and median PM of ΔPQR (see Fig 6.41). Show that ΔABC ~ ΔPQR.

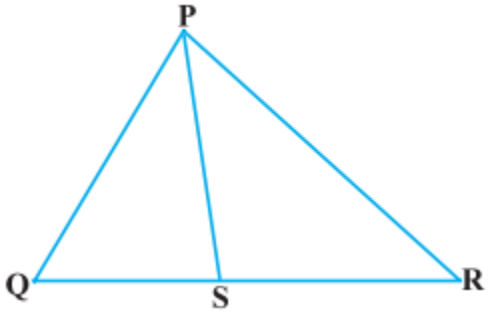


15. D is a point on the side BC of a triangle ABC such that ∠ADC = ∠BAC. Show that $CA^2 = CB.CD$

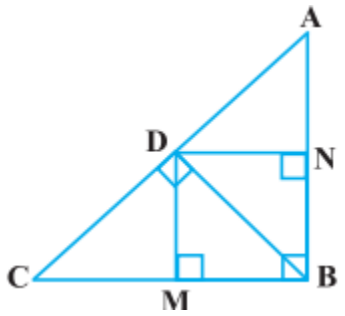
16. Sides AB and AC and median AD of a triangle ABC are respectively proportional to sides PQ and PR and median PM of another triangle PQR. Show that ΔABC ~ ΔPQR.

17. A vertical pole of a length 6 m casts a shadow 4m long on the ground and at the same time a tower casts a shadow 28 m long. Find the height of the tower.

18. In Figure, PS is the bisector of ∠ QPR of Δ PQR. Prove that QS/PQ = SR/PR



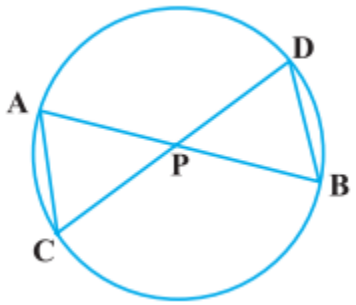
19. In Fig. 6.57, D is a point on hypotenuse AC of ΔABC, such that BD ⊥AC, DM ⊥ BC and DN ⊥ AB. Prove that: (i) $DM^2 = DN . MC$ (ii) $DN^2 = DM . AN$.



20. In Figure, two chords AB and CD intersect each other at the point P. Prove that :

(i) $\Delta APC \sim \Delta DPB$

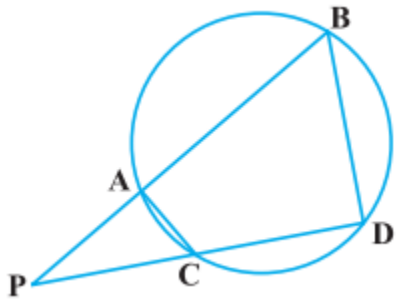
(ii) $AP \cdot PB = CP \cdot DP$



21. In Fig. 6.62, two chords AB and CD of a circle intersect each other at the point P (when produced) outside the circle. Prove that:

(i) $\Delta PAC \sim \Delta PDB$

(ii) $PA \cdot PB = PC \cdot PD$.



22. If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that the other two sides are divided in the same ratio.